

STAT 102 - Important Code for Week 6 (Simulation-Based Hypothesis Tests)

*** NOTE:** These notes are for sample proportion. Adjust as necessary for mean, diff in props, diff in means, and correlation.

LaTeX/Special Characters

- $\hat{p}$ - `\hat{p}`
- $\hat{p}_1$ - `\hat{p}_{1}`
- $\bar{x}$ - `\bar{x}`
- β - `\beta`
- ρ - `\rho`
- μ - `\mu`
- H_0 - `H_{0}`
- $\neq$ - `\neq`

Functions for Hypothesis Tests (for Sample Proportion)

Set **seed** to save randomly-generated information over time (put this at top of **r-chunk**)

- `set.seed(---)`
 - `set.seed(1)`

Construct data frame of sample results for a **sample proportion**

- `DATA-FRAME <- data.frame(CAT-RESPONSE-VAR = c(rep("CAT-1", VALUE-1), rep("CAT-2", VALUE-2)))`
 - `esp <- data.frame(guess = c(rep("correct", 106), rep("incorrect", 329 - 106)))`

Generate **null distribution** for a **sample proportion**

- `NULL-DIST <- DATA-FRAME %>%`
 - `specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%`
 - `hypothesize(null = "point"1, p = NULL-VALUE) %>%`
 - `generate(reps = 1000, type = "draw"2) %>%`
 - `calculate(stat = "prop"3)`
- `null_dist <- esp %>%`
 - `specify(response = guess, success = "correct") %>%`
 - `hypothesize(null = "point", p = 0.25) %>%`
 - `generate(reps = 1000, type = "draw") %>%`
 - `calculate(stat = "prop")`

¹ "point" for 1 variable, "independence" for 2 variables

² "draw" for prop, "bootstrap" for mean, "permute" for diff in means and for diff in proportions

³ "mean", "median", "sum", "sd", "prop", "count", "diff in means", "diff in medians", "diff in props", "correlation"

Compute **observed test statistic** for a **sample proportion**

```
- TEST-STATISTIC <- DATA-FRAME %>%  
  specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%  
  calculate(stat = "prop")  
  - test_stat <- esp %>%  
    specify(response = guess, success = "correct") %>%  
    calculate(stat = "prop")
```

Compute **p-value**

```
- P-VALUE <- NULL-DIST %>%  
  get_p_value(obs_stat = TEST-STAT, direction = "DIRECTION-OF-TEST"4)  
  - p_value <- null_dist %>%  
    get_p_value(obs_stat = test_stat, direction = "greater")
```

Functions for Computing Power (for Sample Proportion)

Step #1: Construct a dataframe and generate a **null distribution**

```
- DATA-FRAME <- data.frame(CAT-RESPONSE-VAR = c(rep("CAT-1", VALUE-1),  
  rep("CAT-2", VALUE-2)))  
  - dat <- data.frame(at_bats = c(rep("hit", 80), rep("miss", 20)))  
- NULL-DIST <- DATA-FRAME %>%  
  specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%  
  hypothesize(null = "point", p = NULL-VALUE) %>%  
  generate(reps = 1000, type = "draw") %>%  
  calculate(stat = "prop")  
  - null <- dat %>%  
    specify(response = at_bats, success = "hit") %>%  
    hypothesize(null = "point", p = 0.25) %>%  
    generate(reps = 1000, type = "draw") %>%  
    calculate(stat = "prop")
```

Step #2: Determine the **critical value(s)**

```
- quantile(NULL-DIST$NULL-VALUE-IN-NULL-DIST, ONE-MINUS-ALPHA)  
  - quantile(null$stat, 0.95)
```

Step #3: Generate an **alternative distribution**

```
- ALT-DIST <- DATA-FRAME %>%  
  specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%  
  hypothesize(null = "point", p = ALT-VALUE) %>%
```

⁴ "less", "greater", "two-sided"

```

generate(reps = 1000, type = "draw") %>%
calculate(stat = "prop")
- alt <- dat %>%
  specify(response = at_bats, success = "hit") %>%
  hypothesize(null = "point", p = 0.33) %>%
  generate(reps = 1000, type = "draw") %>%
  calculate(stat = "prop")

```

Step #4: Find probability of **critical value(s)** (or more extreme) under **alternative distribution**

```

- ALT-DIST %>%
  summarize(power = mean(ALT-VALUE-IN-ALT-DIST >
    quantile(NULL-DIST$ALT-VALUE-IN-ALT-DIST, 0.95)))
- alt %>%
  summarize(power = mean(stat > quantile(null$stat, 0.95)))

```